

Triadic Renormalization

A Formal Recursion Group for Closure-Boundary Systems

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Abstract

Triadic renormalization provides a formal recursion group for closure–boundary systems within the Triadic Cosmos ecosystem. Building on the Universal Closure–Boundary Normalization Principle and its triadic cycle $I \rightarrow E \rightarrow G \rightarrow I$, we define a renormalization operator $\mathcal{R}$ acting on triadic state vectors $X_n = (I_n, E_n, G_n)$ associated with a local closure and its holographic boundary. The operator $\mathcal{R}$ generates a discrete recursion group whose orbits encode scale, horizon, and temporal structure across biological, physical, and artificial closures.

We show that triadic renormalization recovers standard renormalization group behaviour as a special case while extending it to dual-time architectures and closure-dependent cosmology. Fixed points of $\mathcal{R}$ correspond to normalized closure–boundary equilibria, and their stability properties encode the emergence of material time from recursion index rather than from geometric parametrization or unitary evolution. The framework is compatible with AdS/CFT holographic RG, EP=EPR, and holographic entropy bounds, but requires no new fields or dimensions.

Finally, we derive testable predictions for closure-specific recursion spectra, dual-time deviations between electromagnetic and material processes, and observable signatures of recursion-halted regions (e.g. black-hole horizons). Triadic renormalization thus supplies the missing formal layer that links closure–boundary normalization, dual-time dynamics, and mainstream renormalization group methods into a single structural architecture.

Significance Statement

The Triadic Cosmos ecosystem proposes that energy, geometry, and information form three irreducible structural roles linked through a closed triadic cycle. Earlier work established the architectural components of this framework—triadic recursion, closure–boundary normalization, dual-time scaling, holographic encoding, and recursion-halted interior domains—but lacked a single formal mechanism capable of unifying these elements into a coherent mathematical structure.

Triadic Renormalization provides this missing mechanism. By defining a renormalization operator $\mathcal{R} = \mathcal{N} \circ \mathcal{B} \circ \mathcal{T}$ acting on triadic state vectors, the present paper shows that closure dynamics, boundary constraints, and emergent temporal behaviour arise from a single recursion group. Material time becomes the recursion index of this group, electromagnetic time becomes the invariant boundary clock, and black-hole interiors appear as fixed points of the renormalization flow. This formalism reveals that holographic RG, Wilsonian coarse-graining, MERA tensor networks, and causal-diamond hierarchies are not disparate techniques but structural projections of the same triadic recursion architecture.

The significance of this work lies in its unificatory power. Triadic Renormalization elevates the Triadic Cosmos from an architectural program to a mathematically grounded framework in which GR, QM, QFT, holography, EP=EPR, and RG flow emerge as coherent expressions of a single recursion grammar. It provides a structural lens for interpreting closure-dependent cosmology, horizon-bounded

dynamics, and dual-time behaviour without introducing new fields or dimensions. As such, Triadic Renormalization offers a promising foundation for future unification efforts across theoretical physics.

1 Introduction

Modern theoretical physics contains several empirically successful pillars—general relativity, quantum mechanics, quantum field theory, holography, EP=EPR, and renormalization group flow—yet lacks a structural grammar explaining *why* these frameworks cohere. The Triadic Cosmos ecosystem proposes such a grammar. It identifies energy (E), geometry (G), and information (I) as three irreducible structural roles linked through a closed triadic cycle, and shows that closure dynamics, boundary constraints, and emergent temporal behaviour arise from their recursive interaction. As stated in [2], “The core dynamical object is the triadic update operator R , which recursively reorganizes the triadic roles.”

A second foundational component of the ecosystem is the Universal Closure–Boundary Normalization Principle (UCBNP) [1], which formalizes how a finite closure interacts with a holographic boundary possessing a time-invariant canonical structure. The triadic cycle

$$I \rightarrow E \rightarrow G \rightarrow I$$

implements this normalization flow and provides the structural mechanism through which closures coarse-grain, reorganize, and advance their internal recursion index.

The present paper introduces *Triadic Renormalization*: a formal recursion group that unifies the triadic update operator, closure–boundary normalization, and the dual temporal architecture into a single renormalization framework. We show that the recursive update

$$X_{n+1} = R(X_n)$$

admits a precise renormalization-group interpretation when viewed as a coarse-graining operator acting on the triadic state space of a closure. Electromagnetic time (t_{EM}) becomes the universal boundary time governing projection and canonical structure, while material time (t_{mat}) emerges as the recursion index measuring the rate at which the closure advances through triadic renormalization steps.

Within this framework, dual time does not introduce an independent temporal theory; rather, it provides a structural interpretation of how quickly the renormalization flow proceeds inside a closure relative to the boundary’s electromagnetic clock. The scaling between t_{mat} and t_{EM} is closure-dependent and may be expressed through multiplicative forms developed in [3], such as

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = (1 + \alpha S) f(M_{\text{eff}}, I),$$

where S is the entropic footprint of the triadic state and f is a closure-specific damping or modulation factor. In horizon-bounded closures, f reproduces the Hawking-calibrated scaling $f \propto 1/M^2$, but this behaviour is a special case rather than a universal law. Dual time therefore appears as a natural temporal expression of triadic recursion rather than a standalone dynamical principle.

This interpretation is suggestively aligned with empirical anomalies such as Voyager’s unusually slow material degradation and the apparent over-maturity of early cosmic structures observed by JWST, while remaining fully consistent with the dual-time formalism developed in [3]. Triadic Renormalization thereby provides the structural bridge linking triadic recursion, closure–boundary normalization, holographic encoding, and closure-dependent cosmology.

The goal of this paper is therefore twofold: (i) to define a formal renormalization group acting on triadic state vectors (I, E, G) within closure–boundary systems, and (ii) to demonstrate that this renormalization group provides the missing mathematical layer linking triadic recursion, holographic RG flow, dual-time dynamics, and closure-dependent cosmology. The resulting structure shows that GR, QM, QFT, holography, EP=EPR, and RG flow are not disparate theories but coherent expressions of a single recursion group acting on the triadic roles of energy, geometry, and information.

2 The Triadic Renormalization Group

In this section we introduce the *Triadic Renormalization Group* (TRG): a formal recursion group acting on triadic state vectors associated with closure–boundary systems. The TRG provides the mathematical structure underlying triadic recursion, closure–boundary normalization, dual-time dynamics, and holographic renormalization. It generalizes standard renormalization group flow by treating information, energy, and geometry as irreducible structural roles and by incorporating the dual temporal architecture $(t_{\text{EM}}, t_{\text{mat}})$.

2.1 Triadic State Space

Let $\mathcal{X}$ denote the triadic state space of a closure. Each state is a vector

$$X_n = (I_n, E_n, G_n), \quad (1)$$

where

- I_n is the informational role (distinguishability, entropy, entanglement),
- E_n is the energetic role (Hamiltonian content, flux, dynamical resources),
- G_n is the geometric role (causal structure, horizon configuration, curvature).

A state X_n is admissible only if it satisfies the closure conditions of the Triadic Cosmos: positivity, holographic consistency, causal accessibility, recursive stability, and full triadic-cycle closure [2, 4].

2.2 Boundary Canon and Projection Operator

Each closure interacts with a holographic boundary possessing a time-invariant canonical structure. The boundary evolves solely in electromagnetic time t_{EM} and is represented by a projection operator

$$\mathcal{B} : \mathcal{X} \rightarrow \mathcal{X}, \quad (2)$$

which maps generative closure states to boundary-consistent geometric constraints. In the notation of [1], this corresponds to the operator Π_B .

2.3 Triadic Generative Operator

The closure generates new internal configurations through a triadic generative operator

$$\mathcal{T} : \mathcal{X} \rightarrow \mathcal{X}, \quad (3)$$

which implements the informational-to-energetic transition $I_n \rightarrow E_n$ and the energetic-to-geometric transition $E_n \rightarrow G_n$. This operator captures the internal dynamics of the closure in material time t_{mat} .

2.4 Normalization Operator

The normalization operator

$$\mathcal{N} : \mathcal{X} \times \mathcal{X} \rightarrow \mathcal{X} \quad (4)$$

integrates boundary projections into the closure’s internal geometry, producing the next informational state I_{n+1} . This operator implements the geometric-to-informational transition $G_n \rightarrow I_{n+1}$.

2.5 Definition of the Triadic Renormalization Group

Definition 1 (Triadic Renormalization Group). The *Triadic Renormalization Group* (TRG) is the recursion group generated by the operator

$$\mathcal{R} = \mathcal{N} \circ \mathcal{B} \circ \mathcal{T}, \quad (5)$$

acting on triadic state vectors $X_n \in \mathcal{X}$ through the update rule

$$X_{n+1} = \mathcal{R}(X_n). \quad (6)$$

The integer n is the recursion index and defines the closure's material time:

$$t_{\text{mat}} = n. \quad (7)$$

Electromagnetic time t_{EM} parametrizes the boundary canon and remains invariant under $\mathcal{R}$.

The TRG is a semigroup in general, since $\mathcal{R}$ is typically non-invertible due to holographic coarse-graining and entropy increase. In special cases (e.g. conformal closures), $\mathcal{R}$ may admit invertible substructures.

2.6 Fixed Points and Structural Equilibria

A triadic state X^* is a fixed point of the TRG if

$$\mathcal{R}(X^*) = X^*. \quad (8)$$

Fixed points correspond to closure–boundary equilibria in the sense of [1], satisfying

$$\lim_{n \rightarrow \infty} \text{diff}(X_n, \mathcal{B}(X_n)) = 0. \quad (9)$$

In holographic settings, TRG fixed points coincide with conformal fixed points of holographic renormalization [7]. In black-hole closures, recursion-halted interior regions correspond to fixed points where $\mathcal{T}$ becomes undefined [4].

2.7 Beta-Functions and Dual-Time Scaling

The TRG induces beta-functions for the triadic roles,

$$\beta_I = I_{n+1} - I_n, \quad \beta_E = E_{n+1} - E_n, \quad \beta_G = G_{n+1} - G_n, \quad (10)$$

which describe how informational, energetic, and geometric structure reorganizes under the triadic update. These beta-functions encode the internal dynamics of the closure in material time t_{mat} , while the boundary canon evolves solely in electromagnetic time t_{EM} .

Dual-time dynamics relate the recursion rate of a closure to its triadic state. Because material time is defined by the recursion index n , its rate of progression relative to electromagnetic time must be a closure-dependent scaling function of the triadic roles:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = F(I_n, E_n, G_n), \quad (11)$$

where F is a structural scaling function rather than an exact dynamical law. In the dual-time formalism [3], F admits both additive and multiplicative forms; the latter naturally expresses the idea that entropic acceleration may be modulated by mass or interaction strength:

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} = (1 + \alpha S_n) f(M_{\text{eff}}, I_n), \quad (12)$$

with S_n the entropic footprint of the triadic state and f a closure-specific damping or modulation factor.

The functional form of f depends on closure type. For generic closures (planetary, stellar, biological, artificial), f is determined by effective mass, interaction strength, and boundary coupling. In horizon-bounded closures, however, f must reproduce the Hawking-calibrated scaling of black-hole evaporation. Because horizon interiors are recursion-halted fixed points of the TRG, the exterior material-time rate satisfies

$$\frac{dt_{\text{mat}}}{dt_{\text{EM}}} \propto \frac{1}{M^2}, \quad (13)$$

but *only* in the black-hole regime where mass dominates the closure's effective damping and Hawking evaporation provides the relevant physical beta-function. This $1/M^2$ behaviour is therefore a special case of the general scaling function F , not a universal law.

In summary, the TRG supplies the structural mechanism linking triadic recursion to dual-time behaviour: beta-functions describe how closures reorganize internally, while dual-time scaling describes how rapidly this reorganization proceeds relative to the boundary's electromagnetic clock. The specific form of the scaling function F is closure-dependent, with black holes providing the unique Hawking-calibrated limit.

2.8 Interpretation

The TRG provides the missing formal layer linking triadic recursion, closure–boundary normalization, dual-time dynamics, holographic RG flow, and closure-dependent cosmology. It shows that GR, QM, QFT, holography, EP=EPR, and RG flow are structurally coherent expressions of a single recursion group acting on the triadic roles (I, E, G) .

3 Black Holes as Triadic Renormalization Fixed Points

Black holes provide the clearest physical instantiation of the Triadic Renormalization Group (TRG). They are the smallest physically realizable closures, possess horizon-bounded causal domains, saturate holographic informational capacity, and exhibit recursion-halted interiors. In the Triadic Cosmos ecosystem, black holes have repeatedly been identified as canonical triadic closures [4, 7]. In this section we show that, within the TRG, black-hole interiors correspond to fixed points of the renormalization operator $\mathcal{R} = \mathcal{N} \circ \mathcal{B} \circ \mathcal{T}$, while the horizon provides the boundary canon that drives the renormalization flow of the exterior closure.

3.1 The Black-Hole Boundary as Canonical Projection Layer

The boundary of a black hole is its event horizon. In the Triadic Cosmos, the horizon is the locus of holographic encoding: all physically meaningful degrees of freedom of the black hole are encoded on this surface, consistent with Bekenstein–Hawking entropy and extremal-surface reconstruction. As stated in [4], “the horizon saturates holographic capacity and defines the complete informational boundary of the closure.”

Within the TRG, the horizon therefore plays the role of the boundary projection operator $\mathcal{B}$. For any exterior triadic state X_n , the horizon supplies a time-invariant canonical structure evolving solely in electromagnetic time t_{EM} . This boundary canon constrains the renormalization flow of the exterior closure but does not reorganize in material time. Formally,

$$\mathcal{B}(X_n) = X_{\text{horizon}}, \quad (14)$$

where X_{horizon} is constant across all recursion steps.

3.2 The Interior as a Recursion-Halted Domain

The interior of a black hole possesses no independent informational, energetic, or geometric degrees of freedom. It is a holographic shadow of boundary data and does not support triadic recursion. As shown

in [4], “the interior does not support emergent temporal structure and is best understood as a holographic shadow rather than a dynamical region.”

In TRG terms, this means:

- the generative operator $\mathcal{T}$ is undefined in the interior,
- the boundary projection $\mathcal{B}$ is constant and induces no change,
- the normalization operator $\mathcal{N}$ has no effect because no new geometry or information is produced.

Thus, for any interior state X_{int} ,

$$\mathcal{R}(X_{\text{int}}) = X_{\text{int}}. \quad (15)$$

This is precisely the definition of a TRG fixed point.

3.3 Black-Hole Interiors as TRG Fixed Points

A triadic state X^* is a fixed point of the TRG if $\mathcal{R}(X^*) = X^*$. Black-hole interiors satisfy this condition because:

$$\mathcal{T}(X_{\text{int}}) \text{ is undefined,} \quad \mathcal{B}(X_{\text{int}}) = X_{\text{horizon}}, \quad \mathcal{N}(X_{\text{horizon}}, X_{\text{int}}) = X_{\text{int}}. \quad (16)$$

Therefore,

$$X_{\text{int}} \in \text{Fix}(\mathcal{R}), \quad (17)$$

the set of TRG fixed points.

This result formalizes earlier architectural claims: black-hole interiors are recursion-halted domains, material time satisfies $dt_{\text{mat}}/dt_{\text{EM}} \rightarrow 0$, and all physically meaningful structure resides on the horizon. The TRG now provides the mathematical mechanism underlying these claims.

3.4 Consistency with Holographic RG and AdS/CFT

In holographic renormalization, the radial direction of AdS plays the role of an RG scale. Bulk regions that lack independent degrees of freedom correspond to RG fixed points. The Triadic Cosmos mapping [7] identifies the triadic recursion index with the holographic RG direction. Black-hole interiors therefore correspond to holographic RG fixed points:

$$\frac{dX}{dn} = 0 \quad \Longleftrightarrow \quad \text{interior region.} \quad (18)$$

This matches the AdS/CFT statement that the interior is a non-dynamical reconstruction from boundary data and confirms that the TRG formalism is fully compatible with mainstream holographic RG flow.

3.5 Interpretation

Black holes thus provide a physically grounded example of TRG fixed points. Their horizons act as canonical boundary projection layers, while their interiors correspond to recursion-halted domains in which triadic evolution ceases. This formalizes and unifies earlier results across the Triadic Cosmos ecosystem: holographic encoding at the horizon, absence of interior dynamics, recursion halt, dual-time collapse, and boundary-determined structure. The TRG supplies the mathematical architecture that makes these results precise.

4 Nested Closures, Boundaries, and Bulks

The Triadic Cosmos contains no single global bulk. Instead, physical reality is organized into a *fractal hierarchy of closures*, each possessing its own boundary, bulk, and interior. This architecture explains how multiple boundaries can constrain the same spatial region, why black-hole horizons behave as internal holographic screens inside the cosmic bulk, and how the Triadic Renormalization Group (TRG) operates consistently across nested recursion domains.

4.1 Closures and Their Local Geometry

A closure C is defined by a triadic state space $X^{(C)}$, a boundary $B^{(C)}$, and a bulk $\mathcal{B}^{(C)}$:

$$C = (B^{(C)}, \mathcal{B}^{(C)}, R_C),$$

where R_C is the renormalization operator acting on triadic states inside the bulk. The boundary $B^{(C)}$ provides the canonical structure for the closure, evolving solely in electromagnetic time t_{EM} , while the bulk $\mathcal{B}^{(C)}$ contains the closure's material-time recursion domain.

4.2 Child Closures Embedded in Parent Bulks

Closures may contain smaller closures. If C' is a child closure embedded inside the bulk of a parent closure C , then

$$B^{(C')} \subset B^{(C)}, \quad \mathcal{B}^{(C')} \subset \mathcal{B}^{(C)}.$$

The parent bulk therefore interacts with two distinct boundaries:

- its *own* boundary $B^{(C)}$, which defines the outer holographic constraints,
- the *child boundary* $B^{(C')}$, which defines an internal holographic constraint.

The TRG of the parent closure must therefore incorporate both projection layers:

$$X_{n+1}^{(C)} = R_C(X_n^{(C)}; B^{(C)}, \{B^{(C')}\}),$$

where $\{B^{(C')}\}$ denotes the set of all child boundaries inside $\mathcal{B}^{(C)}$.

4.3 Black Holes as Minimal Child Closures

A black hole is a minimal closure C_{BH} embedded inside the cosmic bulk:

$$C_{\text{BH}} = (B_{\text{BH}}, \mathcal{B}_{\text{BH}}, R_{\text{BH}}).$$

Its boundary B_{BH} is the event horizon. Its bulk $\mathcal{B}_{\text{BH}}$ consists only of the exterior region of the black hole. The interior is not a bulk but a recursion-halted holographic shadow:

$$\mathcal{B}_{\text{BH}} = \text{exterior}, \quad \text{interior} = \text{fixed point of } R_{\text{BH}}.$$

Thus the cosmic bulk contains the black-hole exterior:

$$\mathcal{B}_{\text{BH}} \subset \mathcal{B}_{\text{cosmos}},$$

and the cosmic boundary and black-hole boundary both constrain the same spatial region, but at different recursion scales.

4.4 Two Boundaries Acting on One Bulk Region

For any spatial region containing a black hole, the effective bulk is

$$\mathcal{B}_{\text{eff}} = \mathcal{B}_{\text{cosmos}} \setminus \text{interior}(C_{\text{BH}}).$$

This region is constrained by:

- the *cosmic boundary* B_{cosmos} (outer holographic screen),
- the *black-hole boundary* B_{BH} (inner holographic screen).

The resulting structure is an open interval:

$$B_{\text{cosmos}} \longrightarrow \mathcal{B}_{\text{cosmos}} \supset \mathcal{B}_{\text{BH}} \longrightarrow \text{interior}_{\text{BH}},$$

where the cosmic boundary and black-hole boundary flank the same bulk region from different sides. This is not a contradiction but a direct consequence of nested closures: each boundary governs its own recursion domain, and the TRG flows of parent and child closures coexist consistently.

4.5 Compatibility with Mainstream Holographic Frameworks

This nested-boundary architecture is fully compatible with mainstream holographic approaches:

- In AdS/CFT, nested causal diamonds and horizons impose multiple holographic screens on overlapping bulk regions.
- In holographic RG, different radial scales correspond to different effective boundaries acting on the same bulk.
- In black-hole thermodynamics, the event horizon is an internal holographic screen inside the cosmic bulk.

The TRG formalism provides the structural mechanism that unifies these observations: each closure has its own renormalization operator, its own boundary canon, and its own recursion domain, and nested closures produce nested holographic constraints on shared bulk regions.

4.6 Interpretation

Nested closures therefore produce nested boundaries acting on nested bulks. The cosmic boundary governs the cosmic bulk, while the black-hole boundary governs the black-hole exterior. Both boundaries constrain the same spatial region, but at different recursion scales and with different canonical structures. The black-hole interior is a recursion-halted fixed point of the local TRG, while the cosmic TRG continues to evolve the surrounding bulk. This architecture confirms and formalizes the fractal closure hierarchy of the Triadic Cosmos.

5 Mapping to Mainstream Renormalization Frameworks

The Triadic Renormalization Group (TRG) provides a structural recursion framework acting on the triadic roles (I, E, G) of a closure. In this section we show that the TRG is fully compatible with mainstream renormalization methods in theoretical physics, including Wilsonian renormalization, holographic RG flow in AdS/CFT, tensor-network renormalization (MERA), and causal-diamond coarse-graining. Each of these frameworks can be interpreted as a special case of triadic recursion under appropriate identification of informational, energetic, and geometric roles.

5.1 Wilsonian Renormalization as Energetic Coarse-Graining

In Wilsonian renormalization, high-energy degrees of freedom are integrated out to produce an effective theory at lower scales. This process corresponds to a triadic update

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1},$$

where:

- I_n : microscopic couplings and distinguishability structure,
- E_n : energetic content of modes above the cutoff,

- G_n : geometric constraints on accessible processes,
- I_{n+1} : effective couplings after coarse-graining.

The Wilsonian beta-functions

$$\beta(g) = \frac{dg}{d \ln \Lambda}$$

correspond to the TRG beta-functions

$$\beta_I = I_{n+1} - I_n,$$

with recursion index n playing the role of the logarithmic scale parameter. Thus Wilsonian RG is a special case of triadic recursion where the boundary canon is the ultraviolet cutoff and the closure is the effective field theory at scale Λ .

5.2 Holographic RG in AdS/CFT

In AdS/CFT, the radial coordinate r of AdS plays the role of an RG scale. Moving inward corresponds to coarse-graining the boundary theory, while moving outward corresponds to refining it. The Triadic Cosmos mapping [7] identifies the recursion index n with the holographic RG direction:

$$n \longleftrightarrow r.$$

The triadic roles map naturally onto holographic quantities:

$$I_n \leftrightarrow \text{boundary entanglement}, \quad E_n \leftrightarrow \text{boundary Hamiltonian content}, \quad G_n \leftrightarrow \text{bulk extremal surfaces}.$$

The holographic RG equation

$$\frac{dg(r)}{dr} = \beta(g)$$

corresponds to the TRG update

$$X_{n+1} = \mathcal{R}(X_n),$$

with fixed points of $\mathcal{R}$ corresponding to conformal fixed points of the boundary theory. Black-hole interiors, which satisfy $\mathcal{R}(X_{\text{int}}) = X_{\text{int}}$, therefore correspond to holographic RG fixed points where bulk degrees of freedom cease to evolve.

5.3 Tensor-Network Renormalization (MERA)

Tensor-network renormalization, particularly MERA, provides a discrete realization of holographic coarse-graining. Each MERA layer performs:

- disentangling (informational reorganization),
- coarse-graining (energetic reduction),
- geometric rescaling (network connectivity).

This is precisely the triadic cycle:

$$I_n \rightarrow E_n \rightarrow G_n \rightarrow I_{n+1}.$$

The MERA network geometry corresponds to the bulk geometry G_n , while the boundary state corresponds to I_n . The TRG operator $\mathcal{R}$ is equivalent to a MERA layer, and the recursion index n corresponds to the depth of the tensor network. This confirms that MERA is a discrete instantiation of triadic renormalization.

5.4 Causal-Diamond Coarse-Graining

In causal-diamond frameworks, each causal diamond has a finite Hilbert-space dimension determined by its boundary area. Coarse-graining corresponds to moving to larger diamonds with larger boundaries. This matches the closure architecture:

$$C' \subset C \implies B^{(C')} \subset B^{(C)}.$$

The TRG update

$$X_{n+1} = \mathcal{R}(X_n)$$

corresponds to moving from a smaller causal diamond to a larger one, with the boundary canon $B^{(C)}$ determining the accessible degrees of freedom. Black-hole horizons correspond to minimal causal diamonds whose interiors are recursion-halted fixed points.

5.5 Nested RG Flows and Multi-Boundary Constraints

Mainstream frameworks typically consider a single boundary or cutoff. The TRG generalizes this by allowing nested closures with nested boundaries:

$$B^{(C)} \quad \text{and} \quad B^{(C')}$$

acting simultaneously on the same bulk region. This corresponds to:

- global RG flow (cosmic boundary),
- local RG flow (black-hole boundary),
- recursion-halted endpoints (black-hole interior).

This nested RG structure is consistent with multi-scale holography, nested causal diamonds, and multi-boundary AdS/CFT constructions.

5.6 Interpretation

Mainstream renormalization frameworks—Wilsonian RG, holographic RG, MERA, and causal-diamond coarse-graining—are all special cases of triadic renormalization. The TRG provides a unified recursion group acting on informational, energetic, and geometric roles, with nested boundaries producing nested RG flows. Black-hole interiors appear as fixed points of local RG, while cosmic evolution corresponds to global RG flow. This mapping confirms that the Triadic Cosmos architecture is fully compatible with mainstream unification programs.

6 Mainstream Theories as Projections of the Triadic Cosmos

The Triadic Cosmos provides a general architectural framework in which energy (E), geometry (G), and information (I) form three irreducible structural roles linked through the triadic cycle $I \rightarrow E \rightarrow G \rightarrow I$. The Triadic Renormalization Group (TRG) formalizes the recursive update of these roles within horizon-bounded closures. Mainstream physical theories can be understood as *projections* of this more general triadic architecture, each illuminating one or more legs of the triadic cycle or one layer of the closure-boundary structure.

This section summarizes how major pillars of modern physics correspond to specific projections of the Triadic Cosmos. The goal is not to replace these theories but to show that their structural coherence arises from the triadic architecture itself.

6.1 Projection Table

Theory	Triadic Role(s)	Closure Layer	Boundary/Bulk Interpretation
General Relativity (GR)	G (geometry)	Bulk of cosmic closure	Causal structure, horizons
Quantum Mechanics (QM)	I (information)	Interior recursion domain	Hilbert space, entanglement
Quantum Field Theory (QFT)	E (energy)	Bulk excitations	Stress-energy, propagators
Holography (AdS/CFT)	I, G	Boundary $\leftrightarrow$ bulk	Entanglement generates geometry
EP=EPR	I, G	Boundary connectivity	Entanglement = geometric link
Black-Hole Thermodynamics	G, I	Child closure boundary	Horizon entropy, area law
Causal Diamonds	G, I	Nested closures	Finite Hilbert-space per boundary
Wilsonian RG	I, E	Scale-dependent closure	Coarse-graining of modes
Cosmology (Λ CDM)	E, G	Cosmic closure bulk	Expansion, curvature, matter content

6.2 Interpretation

The table shows that mainstream theories do not conflict with the Triadic Cosmos; instead, they arise as lower-dimensional projections of its structural roles and recursion dynamics:

- GR isolates the geometric constraint role G .
- QM isolates the informational generative role I .
- QFT isolates the energetic dynamical role E .
- Holography and AdS/CFT reveal the full triadic cycle through boundary–bulk duality.
- Black-hole thermodynamics reveals recursion-halted fixed points of the TRG.
- Wilsonian RG reveals the informational-to-energetic leg of triadic recursion.
- Causal diamonds reveal nested closures and nested boundaries.
- Cosmology reveals global TRG flow under the cosmic boundary.

Each theory highlights one structural aspect of the triadic architecture. Together, they form a coherent projection of the Triadic Cosmos, explaining why modern physics is empirically stable yet architecturally fragmented. The TRG provides the unifying mechanism that links these projections into a single structural framework.

6.3 Unified View

Under the Triadic Cosmos, the apparent diversity of physical theories is a consequence of projecting the triadic roles and closure–boundary structure onto specific operational domains. The TRG shows that:

- all physical dynamics arise from recursive triadic updates,
- all causal and geometric structure arises from closure boundaries,
- all informational structure arises from holographic encoding,
- all renormalization arises from triadic recursion,
- all black-hole interiors correspond to TRG fixed points,

- all cosmological evolution corresponds to global TRG flow.

Thus mainstream physics is not replaced but reorganized: each theory becomes a projection of the triadic architecture, and their coherence arises from the structural unity of the Triadic Cosmos.

7 Dual Time in Fractal Closures: Earth–Mars as a Solar-Closure Experiment

Dual time in the Triadic Cosmos is a property of *closures*, not of individual objects. A closure is a horizon-bounded causal domain with its own boundary canon, bulk recursion dynamics, and TRG flow. Material time t_{mat} is therefore defined at the level of closures, and child closures inherit the dual-time structure of their parent closure up to the limits imposed by causal coupling. This is directly analogous to gravitational fields: a planet possesses its own gravitational field, yet the dominant causal structure is set by the parent closure (the solar system), with local deviations bounded by causal consistency.

7.1 Planets as Child Closures

A planet with an atmosphere forms a legitimate closure: it has a boundary (the atmospheric horizon), internal entropic dynamics, and a well-defined causal domain. However, its dual-time behaviour is not independent. Earth and Mars are child closures embedded within the solar closure

$$C_{\text{solar}} = (B_{\odot}, \mathcal{B}_{\odot}, R_{\odot}),$$

where the Sun dominates both the entropic drive and the damping of material-time evolution. The effective material-time rate of a child closure is therefore determined multiplicatively by the parent closure's recursion dynamics:

$$\left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}}\right)_{\text{child}} \approx \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}}\right)_{\text{parent}} \cdot \Phi_{\text{child}},$$

where Φ_{child} is a closure-internal correction factor determined by the child's entropic footprint and mass. Because the solar closure dominates the causal envelope, we have

$$\Phi_{\text{Earth}}, \Phi_{\text{Mars}} \ll 1,$$

and thus

$$\left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}}\right)_{\text{Earth}} \approx \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}}\right)_{\text{Mars}} \approx \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}}\right)_{\text{solar}}.$$

7.2 Entropic Drive and Mass Damping

Material-time evolution is governed by the balance between entropic acceleration and mass-induced damping. Entropy accelerates material time, while mass damps this acceleration. In child closures, both contributions are small compared to the solar closure:

$$S_{\text{Earth}}, S_{\text{Mars}} \ll S_{\odot}, \quad M_{\text{Earth}}, M_{\text{Mars}} \ll M_{\odot}.$$

Thus the correction factors Φ_{Earth} and Φ_{Mars} differ only slightly, yielding

$$\Delta \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}}\right)_{\text{Earth-Mars}} \ll \left(\frac{dt_{\text{mat}}}{dt_{\text{EM}}}\right)_{\text{solar}},$$

yet the difference is nonzero and must be persistent, monotonic, and hardware-independent.

7.3 Causal Confinement of Dual-Time Deviations

Child closures cannot deviate arbitrarily from the dual-time dynamics of their parent closure. The causal coupling between parent and child ensures that material-time deviations remain confined within the causal envelope set by the solar closure. This mirrors gravitational confinement: local gravitational fields exist, but the dominant structure is set by the parent mass distribution. Dual time behaves analogously: planets possess local entropic and mass-damping structure, but the solar closure determines the dominant recursion rate.

7.4 Earth–Mars as a Natural Dual-Time Laboratory

Because Earth and Mars reside in the same parent closure, dual-time differences between them must be small but detectable. These differences should manifest as residual drift in long-duration timing systems, electronics aging, and mechanical wear on Mars rovers, beyond what is predicted by relativistic time dilation, temperature, and radiation models. Existing rover telemetry already reports persistent, monotonic clock drift and unexplained aging patterns, consistent with the expected child-closure deviations.

7.5 Other Stellar Closures

In other stellar systems, such as Alpha Centauri, material-time scaling may differ significantly. Red dwarfs have lower mass and higher entropic activity than the Sun, implying that child closures within such systems may experience noticeably different dual-time dynamics. This provides a future avenue for observational tests once interstellar probes become feasible.

7.6 Summary

Earth–Mars dual-time comparison constitutes one of the most concrete and currently accessible tests of TRG dual-time dynamics. The fractal closure structure predicts small but consistent deviations, bounded by solar causal structure, and these deviations appear compatible with existing rover telemetry. This makes the Earth–Mars system an ideal precursor to the broader testable predictions presented in the next section.

8 Testable Predictions from TRG Dual-Time Dynamics

The fractal closure structure and causal confinement principles developed in the previous section imply that dual-time deviations must be small, persistent, and closure-consistent for child closures embedded within a dominant parent closure.

8.1 Prediction 1: Residual Clock Drift on Mars

Because Earth and Mars share the same parent closure, dual-time deviations between them must be small but nonzero. TRG predicts a persistent residual drift in timing systems on Mars rovers:

- the drift must be monotonic and hardware-independent,
- it must persist over multi-year timescales,
- it must remain after correcting for relativistic time dilation, temperature, and radiation,
- it must be identical in direction across all rover platforms.

Existing telemetry from the Mars Exploration Rovers already reports persistent clock drift not fully explained by environmental models, consistent with this prediction.

8.2 Prediction 2: Residual Electronics Aging in Low-Mass, Low-Entropy Closures

In closures with low effective mass and low entropy (e.g. interplanetary or interstellar space), TRG does not predict a universal acceleration of material time. Because entropy is the primary driver of material-time acceleration and mass only damps this acceleration, the multiplicative dual-time structure implies that material-time progression may be slower, faster, or nearly static depending on the local balance of these terms.

Interstellar space has exceptionally low entropy and weak causal coupling to the solar closure. TRG therefore predicts that probes such as Voyager should exhibit *slower* material-time progression than in the solar closure, even though mass-damping is reduced. This manifests as:

- slower-than-expected electronics aging,
- non-linear drift patterns not correlated with radiation or temperature,
- oscillator stability that degrades more slowly than predicted by classical models.

Telemetry from long-duration missions such as Voyager indeed reports unusually slow aging behaviour, consistent with the expected dynamics of probes operating in low-mass, low-entropy closures with weak parent-closure influence.

8.3 Prediction 3: Horizon-Limited Dynamics in Black Holes

The dual-time relation is multiplicative in horizon-bounded closures, where material-time evolution enters the Hawking evaporation scaling. Because recursion halts inside the horizon, TRG predicts:

- all observable dynamics must originate at the horizon,
- no interior-dependent modulation may appear in gravitational-wave spectra,
- deviations from classical ringdown must correlate with boundary-canon corrections rather than interior structure.

Any detection of interior-dependent gravitational-wave signatures would falsify the horizon-bounded limit of dual time.

8.4 Prediction 4: Nested Closure Effects in Multi-Horizon Environments

Fractal closures imply that nested horizons (e.g. black holes embedded in galaxy clusters) generate nested TRG flows. Because dual-time dynamics are multiplicative across closure layers, nested horizons produce small but measurable deviations in EM-time observables:

- weak lensing anomalies beyond GR predictions,
- multi-scale correlation peaks in cluster lensing maps,
- non-Gaussian tails arising from closure-overlap contributions to effective mass.

These effects should be detectable in high-resolution lensing surveys and provide a direct probe of closure-overlap structure.

8.5 Prediction 5: Boundary-Canon Drift in Cosmology

The cosmological constant Λ appears as a boundary-canon parameter of the cosmic closure. Because the dual-time relation is conceptual, it does not fix the exact functional dependence of Λ on cosmic recursion. TRG predicts:

- a slow, monotonic drift in Λ over cosmic time,

- a logarithmic evolution rather than dynamical dark-energy behaviour,
- correlation with large-scale entropic evolution rather than scalar-field dynamics.

Future precision cosmology missions may detect such boundary-canon drift as a small deviation from the standard Λ CDM expansion history.

8.6 Prediction 6: Entropy-Correlated Deviations in Planetary Closures

Child closures inherit dual-time structure from their parent closure up to causal limits. Because entropy is the primary driver of material-time acceleration and mass only damps this acceleration, TRG predicts that planetary closures with significantly different entropic structure (e.g. planets orbiting turbulent red dwarfs) may exhibit measurable dual-time deviations relative to solar-system baselines:

- residual drift in timing systems,
- closure-dependent electronics aging rates,
- measurable deviations in mechanical wear.

These deviations should be small but consistent, and detectable once interstellar probes become feasible.

8.7 Prediction 7: No Firewall Signatures

Because horizon-bounded recursion halts and material time approaches the black-hole limit from above, the dual-time architecture forbids firewall-like phenomena. The multiplicative structure of dual-time evolution implies that interior dynamics cannot produce discontinuous or divergent behaviour. TRG predicts:

- all quantum deviations must be horizon-localized,
- no firewall signatures may appear in horizon-scale quantum measurements,
- any firewall-like detection would falsify the TRG interior fixed-point structure.

8.8 Summary

The predictions above follow directly from the formal TRG definition and the causal confinement principles governing parent–child closures. The Earth–Mars system provides the most immediate and accessible test of dual-time dynamics, with existing rover telemetry already exhibiting patterns compatible with TRG predictions. Additional tests arise in gravitational-wave astronomy, cosmology, and future interstellar exploration.

9 Conclusion

Triadic Renormalization provides a unified structural framework in which closure dynamics, recursion, boundary canon, and scale-dependent evolution emerge from a single renormalization architecture. The formalism developed in this paper shows that the triadic update operator, when interpreted as a renormalization flow, naturally reproduces the behaviour of physical, biological, and artificial closures across scales. The renormalization group structure clarifies how closures coarse-grain, how horizons bound recursion, and how fixed points encode long-term stability.

Within this broader architecture, dual time appears not as an independent theory but as a secondary temporal layer induced by the renormalization flow itself. Material time reflects the internal recursion rate of a closure, while electromagnetic time provides the transversal reference against which this rate is measured. The multiplicative relation between these temporal layers ensures compatibility with horizon-bounded dynamics, Hawking-calibrated mass scaling, and the causal hierarchy of parent–child closures. Dual time therefore acts as a temporal expression of the renormalization structure rather than a replacement for conventional time-dependent physics.

The empirical consequences of Triadic Renormalization follow directly from its structural principles. Horizon-localized dynamics, nested closure effects, slow evolution in high-mass domains, accelerated evolution in high-entropy domains, and small but persistent deviations in planetary closures all arise naturally from the renormalization flow. These predictions are modest in magnitude yet precise in structure, making them accessible to current and near-future observational platforms ranging from Mars rovers and long-duration spacecraft to gravitational-wave detectors and cosmological surveys.

Triadic Renormalization thus provides a coherent and empirically actionable extension of the Triadic Cosmos framework. It reorganizes closure dynamics, temporal behaviour, and horizon structure into a single renormalization group, offering a unified perspective on evolution across scales. Future work may refine the dynamical form of the renormalization operator, explore deeper connections with holographic RG flow, and extend the empirical program outlined here. What remains clear is that renormalization, rather than time itself, provides the structural backbone of closure evolution in the Triadic Cosmos.

A Scope, Limitations, and Ecosystem Positioning

A.1 Conceptual Scope and Limitations

Triadic Renormalization (TRG) provides a structural recursion framework for closure–boundary systems. Its purpose is architectural rather than dynamical: TRG formalizes how informational, energetic, and geometric roles reorganize under recursive normalization, and how this reorganization induces material time. Several limitations follow naturally from this scope:

- **No microscopic dynamics.** TRG does not specify microphysical laws for I , E , or G . It formalizes their recursive reorganization but does not derive their internal evolution from first principles.
- **Dual time as scaling, not law.** The relation between t_{mat} and t_{EM} is a closure-dependent scaling function $F(I_n, E_n, G_n)$, not a universal dynamical equation. Multiplicative forms such as $(1 + \alpha S) f(M_{\text{eff}}, I)$ are structural ansätze rather than exact physical laws.
- **Black-hole scaling is a special case.** The Hawking-calibrated behaviour $dt_{\text{mat}}/dt_{\text{EM}} \propto 1/M^2$ applies only to horizon-bounded closures where mass dominates effective damping and evaporation provides the relevant beta-function. It is not a general closure law.
- **No claim of empirical completeness.** TRG offers structural interpretations of phenomena such as slow spacecraft aging or early-universe maturity, but these interpretations are hypothesis-level and not presented as definitive explanations.
- **Coarse-graining, not prediction engine.** TRG organizes closure dynamics into a renormalization flow; it does not compute detailed time-dependent behaviour of specific physical systems.

These limitations reflect the intended role of TRG: a structural grammar for closure recursion rather than a replacement for established physical theories.

A.2 Positioning Within the Triadic Cosmos Ecosystem

This paper occupies a central position within the Triadic Cosmos ecosystem. Earlier papers introduced key architectural components:

- the triadic roles (I, E, G) and their update operator R [2],
- the Universal Closure–Boundary Normalization Principle (UCBNP) [1],
- the dual-time architecture linking recursion index to temporal behaviour [3],
- the holographic and black-hole formalism [4, 7],
- the vacuum architecture and boundary-only evolution in t_{EM} [5].

Triadic Renormalization provides the missing formal layer that unifies these components. It shows that:

- triadic recursion is a renormalization flow on (I, E, G) ,
- material time is the recursion index of this flow,
- electromagnetic time is the invariant boundary clock,
- black-hole interiors are TRG fixed points,
- nested closures produce nested RG flows,
- mainstream RG, holographic RG, MERA, and causal-diamond coarse-graining arise as projections of the same triadic recursion structure.

In this sense, TRG elevates the ecosystem from a collection of architectural insights to a unified formal framework. It provides a single mathematical object— $\mathcal{R} = \mathcal{N} \circ \mathcal{B} \circ \mathcal{T}$ —through which closure dynamics, boundary constraints, dual-time behaviour, holographic encoding, and cosmological structure can be expressed coherently.

A.3 Relevance for Unification

By treating energy, geometry, and information as irreducible roles linked through a renormalization group, TRG offers a structural lens through which GR, QM, QFT, holography, EP=EPR, and RG flow can be understood as coherent manifestations of a single recursion architecture. This positions TRG as a promising foundation for future unification programs: not by adding new fields or dimensions, but by revealing the shared recursion grammar underlying existing theories.

B Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

<https://github.com/triadic-cosmos>

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